

SOAL 9 (NPM: 91124082)

Solusi dengan Metode Simpleks

$$Z = 343$$

$$X_1 = 2,14$$

$$X_2 = 1,71$$

JAWABAN**Iterasi 1****Subset 1 (Tidak Digunakan)**

$$\text{Maximize } Z = 80X_1 + 100X_2$$

Subject to

$$4X_1 + 2X_2 \leq 12$$

$$X_1 + 5X_2 \leq 15$$

$$X_2 \geq 2$$

Perhitungan subset 1

- Mencari X_1**

$$4X_1 + 2X_2 = 12$$

$$4X_1 + 2(2) = 12$$

$$4X_1 + 4 = 12$$

$$4X_1 = 8$$

$$X_1 = 2$$

$$X_1 + 5X_2 = 15$$

$$X_1 + 5(2) = 15$$

$$X_1 + 10 = 15$$

$$X_1 = 5$$

- Mencari Z**

$$Z = 80X_1 + 100X_2$$

$$Z = 80(2) + 100(2)$$

$$Z = 160 + 200$$

$$Z = 360 \rightarrow \text{Solusi tidak}$$

digunakan karena melebihi batas

Z pada permasalahan

$$Z = 80X_1 + 100X_2$$

$$Z = 80(5) + 100(2)$$

$$Z = 400 + 200$$

$$Z = 600 \rightarrow \text{Solusi tidak}$$

digunakan karena melebihi batas

Z pada permasalahan

Subset 2

$$\text{Maximize } Z = 80X_1 + 100X_2$$

Subject to

$$4X_1 + 2X_2 \leq 12$$

$$X_1 + 5X_2 \leq 15$$

$$X_2 \leq 1$$

Perhitungan subset 2

- Mencari X_1**

$$4X_1 + 2X_2 = 12$$

$$4X_1 + 2(1) = 12$$

$$4X_1 + 2 = 12$$

$$4X_1 = 10$$

$$X_1 = 2,5$$

$$X_1 + 5X_2 = 15$$

$$X_1 + 5(1) = 15$$

$$X_1 + 5 = 15$$

$$X_1 = 10$$

- Mencari Z**

$$Z = 80X_1 + 100X_2$$

$$Z = 80(2,5) + 100(1)$$

$$Z = 200 + 300$$

$$Z = 300$$

$$Z = 80X_1 + 100X_2$$

$$Z = 80(10) + 100(1)$$

$$Z = 400 + 100$$

$$Z = 500 \rightarrow \text{Solusi tidak}$$

digunakan karena melebihi batas

Z pada permasalahan

Iterasi 2

Subset 3

Maximize $Z = 80X_1 + 100X_2$

Subject to

$$4X_1 + 2X_2 \leq 12$$

$$X_1 + 5X_2 \leq 15$$

$$X_1 \geq 3$$

Perhitungan subset 3

- Mencari X_2

$$4X_1 + 2X_2 = 12$$

$$4(3) + 2X_2 = 12$$

$$12 + 2X_2 = 12$$

$$2X_2 = 0$$

$$X_2 = 0$$

$$X_1 + 5X_2 = 15$$

$$3 + 5X_2 = 15$$

$$5X_2 = 12$$

$$X_2 = \frac{12}{5} = 2,4$$

- Mencari Z

$$Z = 80X_1 + 100X_2$$

$$Z = 80(3) + 100(0)$$

$$Z = 240 \text{ (Calon Solusi)}$$

$$Z = 80X_1 + 100X_2$$

$$Z = 80(3) + 100(2,4)$$

$$Z = 240 + 240$$

$$Z = 480 \rightarrow \text{Solusi tidak}$$

digunakan karena melebihi batas

Z pada permasalahan

Subset 4 (Tidak Digunakan)

Maximize $Z = 80X_1 + 100X_2$

Subject to

$$4X_1 + 2X_2 \leq 12$$

$$X_1 + 5X_2 \leq 15$$

$$X_1 \leq 2$$

Perhitungan subset 4

- Mencari X_2

$$4X_1 + 2X_2 = 12$$

$$4(2) + 2X_2 = 12$$

$$8 + 2X_2 = 12$$

$$2X_2 = 4$$

$$X_2 = 2$$

$$X_1 + 5X_2 = 15$$

$$2 + 5X_2 = 15$$

$$5X_2 = 13$$

$$X_2 = \frac{13}{5} = 2,6$$

- Mencari Z

$$Z = 80X_1 + 100X_2$$

$$Z = 80(2) + 100(2)$$

$$Z = 160 + 200$$

$$Z = 360 \rightarrow \text{Solusi tidak}$$

digunakan karena melebihi batas

Z pada permasalahan

$$Z = 80X_1 + 100X_2$$

$$Z = 80(2) + 100(2,6)$$

$$Z = 160 + 260$$

$$Z = 420 \rightarrow \text{Solusi tidak}$$

digunakan karena melebihi batas

Z pada permasalahan